Efficient Adjacent Feature Harmonizer Network With UAV-CD+ Dataset for Remote Sensing Change Detection

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***Abstract*— Remote sensing change detection (RSCD) aims to identify changes within bi-temporal registered images. However, existing deep learning (DL)-based RSCD networks often suffer from large numbers of parameters, high computational complexity, and low inference speed, making it challenging to achieve efficient inference in real-world deployments. In addition, current models lack robust feature-fitting capabilities, necessitating the development of an efficient and powerful RSCD model to address this issue. Therefore, we propose a novel RSCD network named efficient adjacent feature harmonizer network (EAFH-Net) with fast computational speed and lightweight design. It is based on MobileNetV2, considering that change maps of different sizes contain temporal information of bitemporal features and spatial information at various scales, we introduce a multiscale feature neighbor fusion module (MFNFM) to address the lack of interaction between sophisticated-level and elementary-level features, and spatial and channel feature harmonizer module (SCFHM) to harmonize the spatiotemporal information of the change maps. Moreover, data-driven DL algorithms face another challenge due to insufficient granularity and the need for more practical datasets. Therefore, we present unmanned aerial vehicle (UAV)-CD+, a dataset comprising 2002 pairs of bi-temporal UAV low-altitude images, each sized at 1024 × 1024. We performed experiments on three publicly accessible datasets in conjunction with UAV-CD+, comparing the results with other state-of-the-art (SOTA) methods. EAFH-Net attains the utmost precision, obtaining 91.74% on LEVIR-CD, 84.28% on SYSU-CD, 95.07% on WHU-CD, 79.12% on CLCD, and 70.12% on UAV-CD+. We have our model code available at the following link: https://github.com/yikuizhai/UCSFH-Net.**

***Index Terms*— Multiscale feature neighbor fusion module (MFNFM), remote sensing change detection (RSCD), spatial and channel feature harmonizer module (SCFHM), ultralightweight, unmanned aerial vehicle (UAV) datasets.**

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I. Introduction

REMOTE sensing change detection (RSCD) is a methodology that identifies and segments changing parts within the same geographical area across different time intervals by collecting registered multitemporal remote sensing (RS) images. RSCD is utilized across diverse domains such as urban planning [1], land cover change detection (CD) [2], catastrophe evaluation [3], forest cover change [4], resource management [5], and environmental monitoring [6]. In the past, due to the limitations of RS satellites as data collection platforms, including insufficient spatial resolution that made it difficult to capture very small changes in ground objects, long observation intervals that hindered the collection of high-frequency multitemporal images, and high costs and technical barriers, multitemporal RSCD faced significant challenges. On the other hand, the high spatial resolution, flexible data collection, and low cost of unmanned aerial vehicles (UAVs) make them a promising future mainstream platform for RSCD data collection.

Previously, early CD tasks depended on traditional handcrafted methods for feature extraction, examples of these methods include image differencing (ID) [7], principal component analysis (PCA) [8], and change vector analysis (CVA) [9]. However, achieving high accuracy in CD tasks through traditional handcrafted feature extraction methods proved to be challenging. In recent years, with the rapid advancement of convolutional neural networks (CNNs) and deep learning

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(DL), the robust feature extraction capability of CNNs has garnered significant attention and adoption among researchers. Downstream tasks such as image segmentation [10], [11] and object detection [12], [13] have demonstrated improved performance through CNN-based feature extraction.

Currently, many CD methods rely on feature extraction through CNNs. However, according to experiments and statistics, existing CD models with good accuracy suffer from high computational costs and large model sizes, making it challenging to deploy them on low-computational-edge devices. To facilitate the deployment of CD models on UAVs and save costs associated with transmitting bi-temporal RS images captured by UAVs to end devices for inference, there is a need to develop high-precision, fast-inference, and lightweight models for this task. While previous CD models have shown excellent performance only in bi-temporal images captured by RS satellites, to achieve high-precision CD in both satellite and UAV images, we propose the ultralightweight CD network with comparable parameter count and lower computational costs called efficient adjacent feature harmonizer network (EAFH-Net). Conversely, taking into account the limitations inherent in historical satellite imagery and high-altitude aerial imagery, particularly their low spatial resolution, which hinders the capture of subtle changes on the Earth's surface, we introduce a novel low-altitude UAV aerial CD dataset named UAV-CD+. We will conduct experiments using three publicly available satellite and aerial CD datasets, as well as our proposed UAV-CD+ dataset. The results will demonstrate that our proposed EAFH-Net achieves state-of-the-art (SOTA) performance, offering superior results with reduced parameter count and computational cost. To achieve the lightweight design of the CD network, we utilize MobileNetV2 [14] as the backbone network instead of the ResNet series [15]. The satellite-captured images and UAV-captured images are shown in Fig. 1.

In CD tasks, MobileNetV2 can serve as a lighter contextual encoder compared to ResNet, with performance comparable to or even superior to ResNet. Inspired by multiscale feature fusion methods like feature pyramid networks (FPNs) [16] and pyramid attention networks (PANs) [17], we devised a multiscale feature neighbor fusion module (MFNFM) to incorporate features acquired by the backbone network. This module adeptly merges sophisticated and elementary-levels features extracted by MobileNetV2, strengthening the extracted bi-temporal information and refining granularity to achieve better performance.

In addition to the mentioned encoders, we designed a computationally efficient and lightweight decoder, which includes a spatial and channel feature harmonizer module (SCFHM). The SCFHM acts as a harmonizer for encoding the difference maps. It harmoniously integrates the bi-temporal features of the difference maps across spatial and channel dimensions, making them more prominent in the feature space. By harmonizing both sophisticated and elementary-level features and synergistically blending them across spatial and channel dimensions, it facilitates better integration of the bi-temporal features, thereby presenting the features more comprehensively and enhancing the precision of the CD model's detections.

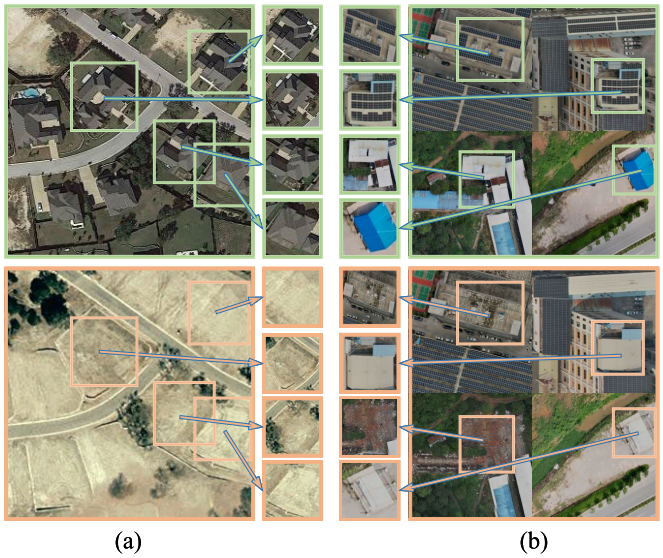


Fig. 1. Samples of two images taken at different points in time. (a) Satellite imagery. (b) UAV imagery.

By combining MFNFM and SCFHM, our proposed network EAFH-Net achieves $F1$-scores of 91.74%, 84.24%, 95.07%, 79.21%, and 70.12% on the LEVIR [18], SYSU [19], WHU-CD [20], CLCD [21], and our proposed UAV-CD dataset UAV-CD+, respectively. Furthermore, EAFH-Net has a small parameter count (4.81M) and extremely low FLOPs (2.93G), making it the CD model with excellent accuracy and minimal FLOPs known to date. Below, our contributions can be encapsulated as follows.

1) We introduce an innovative Siamese-network framework named EAFH-Net, which consists of MFNFM and SCFHM modules. The MFNFM module facilitates the cross-fusion of sophisticated and elementary-levels in proximity to the backbone network, while the SCFHM module harmonizes spatiotemporal feature information of the change map output by the encoder, further amalgamating sophisticated and elementary-levels features to ensure the comprehensiveness of difference map features.

2) Harnessing a lightweight backbone network streamlines the extraction of bi-temporal features. Furthermore, the integration of MFNFM and SCFHM, which embed efficient local convolution processing, enables effective interaction across multiscale features and enriches spatiotemporal information within the change map. This strategy culminates in achieving SOTA performance with a modest parameter count of 4.81M and computational efficiency of 2.93G Flops.

3) We propose UAV-CD+, a UAV-based CD dataset comprising 2002 pairs of 0.06-m resolution RSCD images, each boasting a resolution of 1024 $\times$ 1024 pixels. This dataset covers urban building changes, land changes, water body changes, preconstruction, and postconstruction changes in construction sites, enabling a broader range of CD tasks.

4) We undertake experimentation across four benchmark datasets. Furthermore, we include our devised dataset to ascertain the efficacy and supremacy of the proposed technique. The outcomes demonstrate that EAFH-Net outperforms 12 SOTA models in both accuracy and efficiency.

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The subsequent sections of this article are organized as outlined below. Section II offers a concise overview of relevant research. Section III delves into the intricate nuances of the novel UAV-CD+ dataset. Section IV unveils the architectural framework of the EAFH-Net. Experimental intricacies are elaborated upon in Section V, and finally, conclusions are encapsulated in Section VI.

II. Related Work

A. Traditional CD Methods

Early CD methods traditionally depended on hand-crafted techniques for extracting features from bi-temporal images. For instance, methods like ID [7] involved subtracting images from two different time points to generate a difference map that depicts changes occurring on the Earth's surface. Subsequently, thresholding was applied to segment the difference map to extract and display the changed areas. In addition, PCA [8] utilizes linear transformations to condense the dimensionality of the original data, retaining the maximum data variance in the lower dimensional space. In the context of CD tasks, PCA transformed bi-temporal images into lower dimensional information, with selected thresholds determining principal components and changed components. CVA [9] computed pixel-level differences between bi-temporal images, yielding change vectors comprising the magnitude and direction of change for each pixel between the two-time points. Finally, statistical analysis of these change vectors, along with thresholding, determined significant changes. These methods primarily relied on hand-crafted feature extraction, resulting in lower accuracy and insufficient robustness. Furthermore, in the current landscape, characterized by satellite imagery with exceptional resolution or UAV imagery, these methods encounter challenges in excelling at CD tasks.

B. DL-Driven CD Approaches

With the meteoric progress of DL, many CD methods based on CNNs and transformers [26], [27], [28], [29], [30], [31], have been proposed by researchers. These methods have shown superior accuracy and robustness compared to traditional CD approaches. CNNs are utilized for local feature extraction from bi-temporal images, where sophisticated-level features extracted can locate changed objects, while elementary-level features capture intricate changes within the altered regions. Moreover, modules like FPN and PAN have emerged in the DL field to integrate sophisticated and elementary-level features. Researchers have extensively crafted networks to amalgamate the sophisticated and elementary-level features gleaned by CNNs, aiming to attain heightened precision in CD. For instance, early fully convolutional network (FCN) [32]and its variants are designed for fine-grained CD tasks. Daudt et al. [33] proposed three FCN structures, including the FC encoding and decoding network with skip connections (FC-EF) and two Siamese FCNs (FC-Siam-conc and FC-Siam-diff). Barkur et al. [34] introduced RSCDNet, a pixel-level CD model tailored for bi-temporal high-resolution (HR) RS images. The model integrates the MSA and GL-ASPP blocks. Meanwhile, a plethora of weakly supervised learning and unsupervised learning methods have surfaced. Zhao et al. [35] introduced a CD network named FMCD, informed by feature interplay and multipronged learning. The contextual information features of the FMCD model are acquired through multitier feature interaction modules to extract distinctive features. Wu et al. [36] presented an FCN CD framework employing generative adversarial networks (GANs). This framework amalgamates unsupervised, weakly supervised, region-supervised, and fully supervised CD tasks into a unified structure. Subsequently, utilizing the transformer decoder to reintegrate the enriched tokens into the pixel space, thereby enhancing the original features. However, past CD methods often utilized large-parameter backbones and complex feature fusion modules [37], [38], [39], [40], leading to models with excessive parameters and high computational costs. This made it challenging to deploy them for inference on edge devices, such as UAVs, due to their high computational demands. Therefore, we lean toward designing a CD model with extremely fast computation speed, minimal parameters, and high accuracy.

C. Lightweight CD Methods

In the era of ubiquitous artificial intelligence, where DL models are routinely deployed on mobile devices, researchers increasingly pursue DL models with lower parameter counts and reduced computational costs. This pursuit extends to tasks like RSCD, which also require more efficient lightweight networks.

Li et al. [41] proposed a novel lightweight network named A2Net, which integrates features from adjacent stages through a neighbor aggregation module (NAM). They also introduced a progressive change identifying module (PCIM) to extract temporal difference information from bitemporal features, along with a supervised attention module (SAM) for effective reweighting between high-level and low-level features. Lei et al. [42] introduced an ultralightweight spatial–spectral feature cooperation (SSFC) network named USSFC-Net. The network's strength lies in its multiscale decoupled convolution (MSDConv) design, which flexibly captures multiscale features of changing objects. This design significantly reduces parameter count and computational redundancy. In addition, the network introduces SSFC to enrich its capabilities. Li et al. [43] proposed a lightweight network that uses a focus module for downsampling in the encoding layer. They designed an efficient feature extraction module based on depthwise convolutions by stacking convolutional kernels of different sizes, and introduced a low-parameter multiscale feature fusion module. Jiang et al. [44] proposed a full-scale hybrid network that combines CNN, MLP, and transformer. The MLP helps recover lost global information, while the transformer effectively establishes the association between semantic information and pixel space, enabling a more efficient structure for CD tasks. Recently, many lightweight CD networks have also been proposed [45], [46].

Some of the current lightweight CD models still adopt feature fusion methods with high computational complexity or overly simplistic architectures, making it difficult to ensure

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TABLE I
DIFFERENT CD DATASET INFORMATION

Dataset	Number of Image Pairs	Image Size	Resolution(M)	Modality	Platform/Source
OSCD [22]	24	600×600	10	Multispectral	Sentinel-2 satellites
CDD [23]	16000	256×256	0.03-1	RGB	Google Earth(satellites)
S2Looking [24]	5000	1024×1024	0.5-0.8	RGB	GaoFen,SuperView, BeiJing-2 (satellites)
LEVIR-CD [18]	637	1024×1024	0.5	RGB	Google Earth(satellites)
SYSU-CD [19]	20000	256×256	0.5	RGB	aerial image
WHU-CD [20]	1	32207×15354	0.2	RGB	aerial image
CLCD [21]	600	512×512	0.5-2	RGB	aerial image
DSIFN-CD [25]	3940	512×512	0.03-1	RGB	Google Earth(satellites)
UAV-CD+(ours)	**2002**	**1024×1024**	**0.06**	**RGB**	UAVs

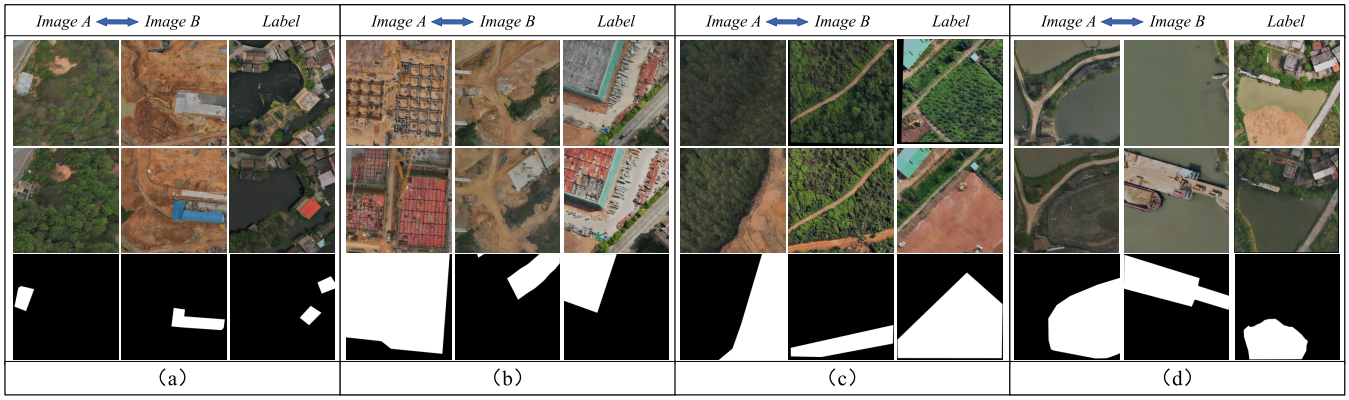


Fig. 2. Examples of captured regions (1024 × 1024) from our UAV-CD+ dataset. Each column represents a category of change: (a) changes in urban architectures, (b) changes in large-scale construction sites, (c) land changes, and (d) changes in water areas.

both model efficiency and performance. With the increasing prevalence of lightweight devices, there is a need for a model that maintains a lightweight structure while achieving high accuracy in CD tasks across various scenarios. Such a model should be deployable on lightweight devices, enabling fast, efficient real-time inference. To ensure the network remains lightweight, we adopt MobileNetV2 as the backbone, with MFNFM for cross-fusion of adjacent sophisticated and elementary-level features after feature extraction. The decoder includes SCFHM which acts as a harmonizer for encoding the difference maps. It harmonizes and integrates bi-temporal features from difference maps across spatial and channel dimensions, enhancing their prominence in the feature space. This harmonization and blending of both sophisticated and elementary-level features facilitates a more comprehensive integration of bi-temporal features, ensuring a more thorough representation.

III. UAV-CD+ DATASET

In the course of CD development, considerable efforts have been dedicated to the creation of CD datasets (CDDs), which play an essential role in advancing the CD field. Table I presents the collected CDDs image pairs, their sizes, resolutions, and modal types, including those captured by satellites [47] and UAVs [48]. For example, the OSCD dataset was established using Sentinel-2 satellite imagery, capturing images at resolutions ranging from 10 to 60 m across 13 bands between ultraviolet and shortwave infrared. This dataset comprises 24 pairs of 600 × 600 pixel images and has been one of the significant datasets in early multispectral CD research.

Following this, in order to facilitate HRCD tasks, researchers introduced the CDD dataset, which involved cropping images from seven pairs of real seasonal change RS images. These images were sized at 4725 × 2700 pixels, downscaled to smaller image pairs formatted in dimensions of 256 × 256 pixels, and spatial resolutions spanning from 0.03 to 1 m. In addition, for instance-level CD tasks, the S2Looking dataset was introduced, comprising 5000 pairs of large-scale oblique satellite images captured at different nonnadir angles. This dataset included over 65 290 annotated change instances from rural areas.

Nowadays, many datasets proposed by researchers are specifically designed for monitoring and detecting urban building changes. The most commonly used datasets include LEVIR-CD [18], SYSU-CD [19], WHU-CD [20], CLCD [21], and DSIFN-CD [25]. Compared to other datasets, our proposed UAV-CD+ dataset features diverse categories of change objects and higher resolutions, aiming to enable CD models to achieve better performance in practical CD applications.

While the datasets mentioned above tackle the issue of CD in multitemporal RS images, it's worth noting that most bi-temporal data are captured by RS satellites or high-altitude aerial platforms. The image resolution and spatial resolution of these datasets are often suitable for CD tasks involving

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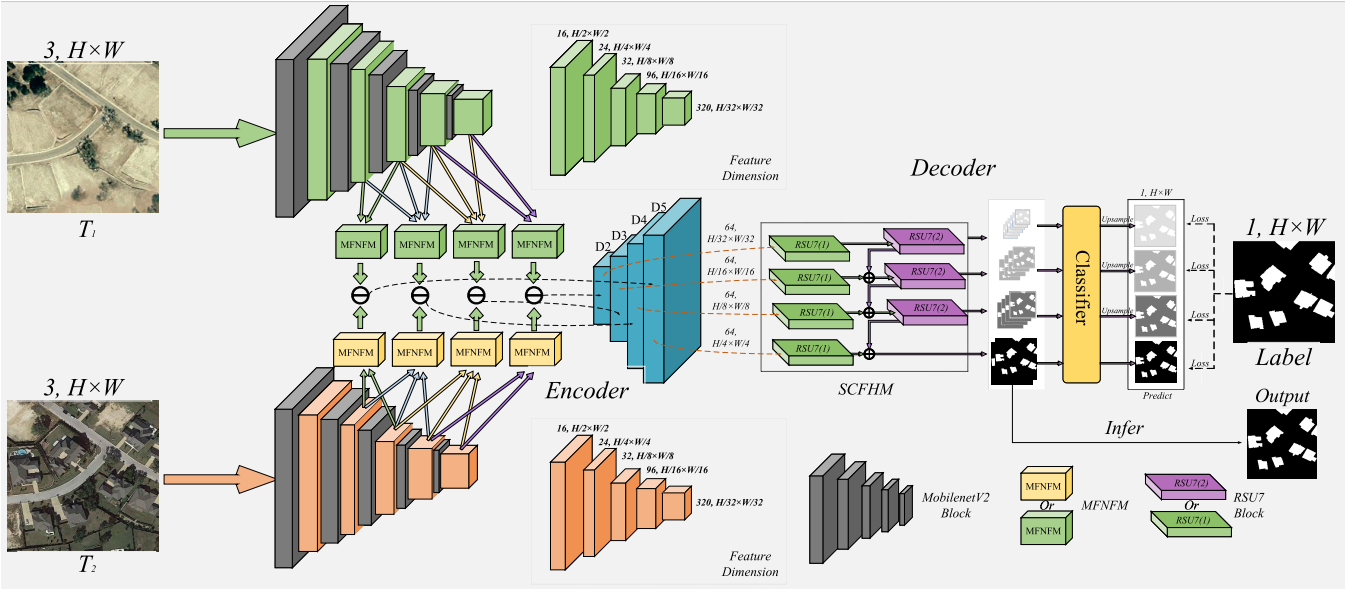


Fig. 3. Proposed EAFH-Net framework utilizes the MobileNetV2 network as the backbone in a Siamese-network configuration with weight sharing, to extract features of bi-temporal image pairs. Features from different scales are fed to the MFNFM module for the spatial information fusion of adjacent features. The output difference map undergoes spatial and channel feature information reconciliation through the SCFHM module, which consists of two stacked residual U-blocks-7 (RSU7) layers. This process further blends sophisticated and elementary-level features. Finally, scale alignment is performed through upsampling, yielding the change map as the output.

significant changes but may not be sufficient for tasks involving subtle changes or small change areas. In addition, many datasets focus exclusively on urban building changes, limiting their applicability to nonbuilding CD tasks and failing to meet the requirements of practical applications.

Our proposed UAV-CD+ dataset utilizes low-altitude drone imagery to complement existing urban building CD datasets. It addresses the need for detecting subtle changes and nonurban building changes, providing a new benchmark for CD and UAV-based CD tasks. The dataset was captured in a prefecture-level city in Guangdong Province and consists of 2002 pairs of images captured by low-altitude drones. Such as other CD datasets, our proposed dataset is suitable for evaluating the performance of CD models in UAV-based CD tasks. Traditional satellite RS images and high-altitude aerial images suffer from issues such as insufficient resolution, difficulty in data acquisition, complex processing procedures, and susceptibility to meteorological conditions. Furthermore, some large-scale RSCD datasets still have room for improvement. For instance, the spatial resolution of certain datasets remains low, and some datasets are relatively small, which may lead to overfitting in data-driven DL techniques. As shown in Table I, the dataset we propose, collected via UAV, has the highest spatial resolution. Moreover, for images of the same size, our dataset surpasses the SYSU-CD, LEVIR-CD, and WHU-CD datasets in terms of the balance between spatial resolution and dataset size. Each image is dimensioned at 1024 × 1024, boasting a spatial resolution of 0.06 m, and there is a two-year interval between captures. Due to the rapid urban development in Guangdong Province, there is a frequent occurrence of illegal construction and unauthorized land occupation. We not only collect data on rapid changes in basic infrastructure, but also gather information on changes in land use, certain large-scale construction sites, and alterations in water bodies. Therefore, our dataset includes a large number of instances of illegal land occupation, captured through low-altitude drone flights to detect instances of small-scale illegal occupation more accurately. We partitioned the initial 2002 pairs of images into training, validation, and test sets, maintaining an 8:1:1 ratio. To address device performance limitations, we cropped the images into numerous pairs, each with a pixel dimension of 256 × 256. During training, we applied random rotation and cropping for data augmentation. The predominant changes in the datasets are visually depicted in Fig. 2, including (a) architectural changes, (b) large-scale construction sites, (c) land alterations, and (d) changes in water areas.

IV. Methodology

In this section, we commence by presenting the proposed model architecture. Subsequently, we offer an intricate breakdown of each constituent of our model. Finally, we introduce the optimization strategy employed for the network model.

A. Overview

Our proposed architecture, depicted in Fig. 3, comprises three components: the backbone network, the MFNFM module, and the decoder. We employ a Siamese network with weight sharing to extract features from bi-temporal images. Prior research has shown that utilizing a backbone network with pretrained parameters aids in model convergence. Therefore, for real-world deployment scenarios, we select MobileNetV2 as the backbone extractor, leveraging pretraining on the ImageNet dataset.

To acquire deep features in CD, the backbone feature extraction network extracts multilayered feature information. These features are subsequently input into the MFNFM for the

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aggregation of multiscale adjacent features. The aggregated features are then passed into the SCFHM module in the decoder to harmonize different channel information.

If we designate image pairs taken at distinct time intervals within the identical location as T_1 and T_2, respectively, and denote the labeled ground truth with change area annotations as L, the comprehensive procedure of EAFH-Net unfolds as outlined below.

1) Initially, the images input T_1 and T_2 undergo feature extraction via the backbone network, which utilizes weight sharing. This results in a collection of multiscale features, denoted as $\{\text{feat}_1^t, \text{feat}_2^t, \text{feat}_3^t, \text{feat}_4^t, \text{feat}_5^t\}$, where t indicates the time points T_1 and T_2.

2) Next, the multiscale features extracted from the shared-weight dual-branch backbone feature extractor are cross-fed into our proposed MFNFM. The features extracted by the dual-branch feature extractor are $\{\text{feat}_2^{t1}, \text{feat}_3^{t1}, \text{feat}_4^{t1}, \text{feat}_5^{t1}\}$ and $\{\text{feat}_2^{t2}, \text{feat}_3^{t2}, \text{feat}_4^{t2}, \text{feat}_5^{t2}\}$. MF$(\cdot)$ refers to the operation of features entering the MFNFM

$$P_n^t = \text{MF}\left(\text{feat}_n^t\right) \quad n \in (2, 3, 4, 5) \quad t \in (1, 2). \tag{1}$$

This process enables the fusion of channels and scales between adjacent features extracted from the backbone feature extractor. Following this, the features from both branches, post MFNFM processing, undergo an absolute difference operation as outlined below:

$$D_i = |p_n^{t1} - p_n^{t2}| \quad n \in (2, 3, 4, 5). \tag{2}$$

3) This operation produces the CD difference maps as shown in 2. Then, the differentially encoded features from both branches are input into the decoder for decoding operations. Then, the features maps D_2, D_3, D_4, D_5 from different scales are input into the SCFHM, resulting in $\{D_2', D_3', D_4', D_5'\}$. Given the necessity to reconcile lower level features with spatial and channel information, the integrated lower level features undergo upsampling and subsequent fusion with adjacent higher level features. This process aims to achieve a better fusion of feature information between different scales, enhancing the change features to acquire more precise change data. Here, SC$(\cdot)$ signifies the feature difference map following the operation process of SCFHM. After the SCFHM self-enhancement of the features of the change map, four high-dimensional change feature maps of different scales are obtained. After passing through the second layer RSU block of SCFHM, these four feature maps undergo dimensionality reduction through a classifier, resulting in four different scale change results with a channel number of 1

$$D_i' = \text{SC}(D_i) \quad i \in (2, 3, 4, 5). \tag{3}$$

4) Finally, the change results from the four scales are upsampled to a pixel size of 256×256, C_5 is the change map output by the inference mode. resulting in change results C_i, where i belongs to $\{2, 3, 4, 5\}$. These change results are then used to compute the final loss (see Section IV-D). The sum of these metrics across all scales promotes more accurate model training

$$C_i = \text{Upsampling}\left(D_i'\right) \quad i \in (2, 3, 4, 5). \tag{4}$$

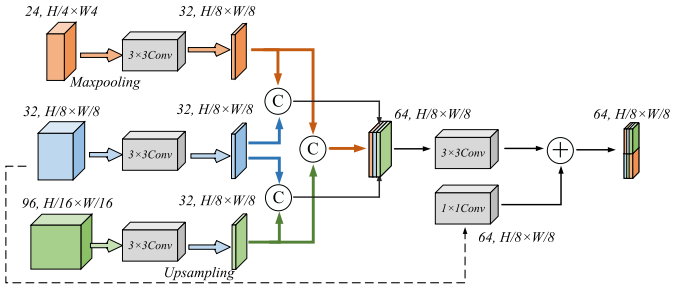


Fig. 4. Structure of the proposed MFNFM.

B. Multiscale Feature Neighbor Fusion Module

Considering the independence of features extracted from the backbone, isolated sophisticated or elementary information cannot simultaneously address both coarse-grained and fine-grained features. This is detrimental to the precise localization and accurate pixel value prediction required for the final CD task. Performing convolution operations on feature maps of varying sizes in a simplistic and independent manner fails to facilitate the integration of features across different levels. Given that each layer's features contain specific representations, it is essential to consider these unique representations within each layer. Therefore, we concatenate adjacent multiscale features in pairs to enable the integration of their specific representations. This approach enhances the completeness of the adjacent multiscale feature representations obtained from MFNFM. Our subsequent experiments demonstrate the effectiveness of this concatenation operation

$$F_2^1 = \text{Conv}_{3\times3}(\text{Maxpooling}(\text{feat}_2)) \tag{5}$$

$$F_3^1 = \text{Conv}_{3\times3}(\text{feat}_3) \tag{6}$$

$$F_4^1 = \text{Upsampling}\{\text{Conv}_{3\times3}(\text{feat}_4)\}. \tag{7}$$

Describing MFNFM in detail, this module is designed to facilitate interaction among adjacent features. For instance, features feat_2, feat_3, and feat_4 extracted from the backbone network undergo the following operations to enrich channel information and refine features: feat_2 is subjected to max-pooling downsampling and a 3×3 convolution to increase channel dimensions, resulting in F_2^1; feat_3 undergoes further feature refinement via a direct 3×3 convolution without altering channel dimensions, yielding F_3^1; finally, feat_4, with high channel dimensions, undergoes channel reduction and upsampling to focus more on fine-grained changes, resulting in F_4^1. Subsequently, the three aligned feature maps $\{F_2^1, F_3^1, F_4^1\}$ are concatenated pairwise along the channel dimension, allowing adjacent features to interact and learn common characteristics, optimizing the edges and textures of change areas. This enhances the network's ability to delineate change areas, resulting in feature maps $\{F_2^2, F_3^2, F_4^2\}$, which are then concatenated along the channel dimension to produce fully integrated features. These features are subsequently dimensionally reduced using a 3×3 convolutional block to acquire the final integrated features

$$F_2^2 = \text{Concat}\left(F_2^1, F_3^1\right) \tag{8}$$

$$F_3^2 = \text{Concat}\left(F_3^1, F_4^1\right) \tag{9}$$

$$F_4^2 = \text{Concat}\left(F_2^1, F_4^1\right). \tag{10}$$

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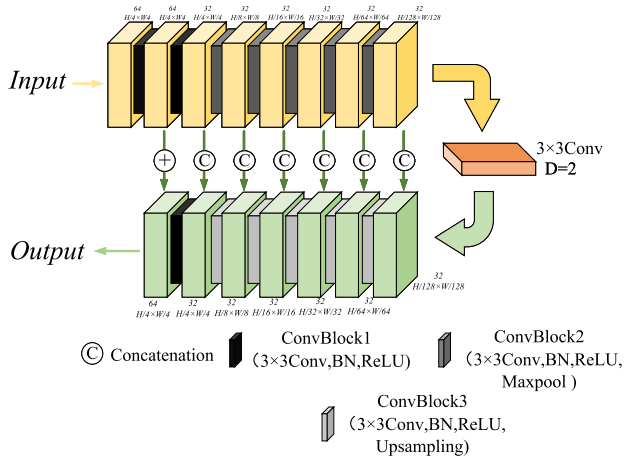


Fig. 5. Overview of residual U-block.

In addition, similar to residual operations, F_3 undergoes channel dimensionality increase through a 1×1 convolution to embed original information into the integrated features, preventing them from deviating too much from the original features. This ensures the model predicts change areas more accurately and with finer detail. The bi-temporal image undergoes this step to obtain $\{P_2(t), P_3(t), P_4(t), P_5(t)\}, t \in (1, 2)$. The specific structure is illustrated in Fig. 4

$$P_3(t) = \mathrm{Conv}_{1\times1}(\mathrm{feat}_3) + \mathrm{Conv}_{3\times3}\left(\mathrm{Concat}\left(F_2^2, F_3^2, F_4^2\right)\right). \tag{11}$$

C. Spatial and Channel Feature Harmonizer Module

Considering that the difference maps D_2, D_3, D_4, D_5 obtained through the Encoder only provide coarse-grained change location information and struggle with precise pixel-level predictions, a deeper feature extraction method is needed to enhance the edge and texture information of the difference maps. To address this, we employ two layers of RSU-blocks [49] in SCFHM to deeply reinforce the precise change information within the difference maps themselves. While RSU-blocks delve into deep information mining of the difference maps, residual operations within the RSU-blocks integrate shallow and deep information of the difference maps, preventing excessive smoothing of fine-grained features. This approach enables simultaneous and accurate prediction of changed regions and their precise changed pixels, the following RSU operations refer to calculations performed through the Residual U-block. The structure is illustrated below

$$D_5'' = \mathrm{RSU}[\mathrm{RSU}(D_5)] \tag{12}$$

$$D_i'' = \mathrm{RSU}\left(D_{i+1}'' + \mathrm{RSU}(D_i)\right) \quad i \in (3, 4) \tag{13}$$

$$D_i'' = D_{i+1}'' + \mathrm{RSU}(D_i) \quad i = 2. \tag{14}$$

The RSU conducts multilayer 3×3 convolutions for feature extraction along with max-pooling downsampling operations to better capture the global features and structural information of the image. In addition, it utilizes a 3×3 convolution block with a dilation of 2 in the final stage to enhance the receptive field for feature extraction while reducing computational overhead, The $\mathrm{DConv}_{3\times3}$ in (16) refers to the above-dilated convolution operation. Following that, via continuous upsampling and concatenation with features from the downsampling stage, sophisticated and elementary-level features are combined to maintain the integrity of feature information. This particular structure is illustrated in Fig. 5

$$D_{2(f)}^{j=7} = \left\{\mathrm{ConvBlock2}[\mathrm{ConvBlock1}(D_2)]^{(k=2)}\right\}^{(l=5)} \tag{15}$$

$$D_{2(b)}^{j=7} = \mathrm{DConv}_{3\times3}\left(D_{2(f)}^{j=7}\right) \tag{16}$$

$$D_{2(b)}^{(j-1)} = \mathrm{ConvBlock3}\left[\mathrm{Concat}\left(D_{2(b)}^{j}, D_{2(f)}^{j}\right)\right] \tag{17}$$

$$D_2' = \mathrm{ConvBlock1}(D_2) + \mathrm{ConvBlock1}\left(D_{2(b)}^{2}\right). \tag{18}$$

The above (15)–(18) represent the computation operations in the RSU, where k in (15) denotes the two-layer ConvBlock1 operations, and l denotes the five-layer ConvBlock2 operations. The specific ConvBlock1, ConvBlock2 and ConvBlock3 operations are shown in Fig. 5. And j represents the level of feature hierarchies in the RSU's forward process, where $3 < j < 8$ and j is an integer; f and b refer to the forward and backward computation operations in the RSU, respectively.

D. Loss Function

The loss function utilized comprises an amalgamation of binary cross-entropy loss (BCELoss) and dice loss [57]. Since CD focuses more on predicting foreground objects, and typically foreground objects occupy a small proportion of the image, dice loss assists in mitigating the adverse effects of foreground–background class imbalance within the samples. That is, in most regions of the image, there are no objects present. Conversely, BCE loss optimizes the model's predictions of changes and nonchanges. Through the amalgamation of BCE loss and dice loss, the model can attain the optimal solution of the objective function under foreground–background class imbalance conditions, thereby facilitating model convergence. In the following formula, y refers to the vector of true binary labels, $\hat{y}$ refers to the vector of predicted probabilities, N refers to the total number of samples, and i refers to the ith training or inference sample. The formulations for BCE loss and dice loss are presented as outlined below

$$L_{\mathrm{BCE}}(y, \hat{y}) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(\hat{y}_i\right) + (1 - y_i)\log\left(1 - \hat{y}\right)\right] \tag{19}$$

$$L_{\mathrm{Dice}}(y, \hat{y}) = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}}{\sum_{i=1}^{N} y_i^2 + \sum_{i=1}^{N} y_i^2} \tag{20}$$

$$L_{\mathrm{total}}(y, \hat{y}) = \sum_{i=2}^{5}\left[L_{\mathrm{Dice}}(y_i, \hat{y}) + L_{\mathrm{BCE}}(y_i, \hat{y})\right]. \tag{21}$$

Algorithm 1 provides an overview of the general workflow of EAFH-Net.

V. Experiments

In this section, we commence by delineating the datasets and benchmark models for comparison utilized in the experiments. Following that, we expound on the detailed

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TABLE II

QUANTITATIVE ANALYSIS OF RSCD MODELS (PRECISION, RECALL, $F1$, AND IOU) ON THREE DATASETS. THE BEST VALUES ARE IN BOLD. ALL RESULTS ARE DESCRIBE AS PERCENTAGES (%)

Dataset	Params(M)	FLOPs(G)	FPS	LEVIR-CD				SYSU-CD				WHU-CD				CLCD			
				Pre	Rec	F1	IoU	Pre	Rec	F1	IoU	Pre	Rec	F1	IoU	Pre	Rec	F1	IoU
FC-Siam-Diff [33]	4.73	1.35	**169.89**	90.52	86.96	88.70	79.70	**91.47**	54.16	68.03	51.56	83.04	76.68	79.74	66.30	69.99	57.67	63.23	46.23
FC-Siam-Conc [33]	5.33	1.55	167.16	88.97	85.41	87.15	77.23	81.84	77.15	79.43	65.87	81.49	72.42	76.69	62.19	64.26	59.06	61.55	44.46
SNUNet [50]	12.03	54.82	45.74	90.06	87.20	88.61	79.55	74.01	81.15	77.41	63.15	84.70	66.80	74.69	59.60	62.42	65.60	63.97	47.03
DSIFN [25]	35.99	79.98	41.44	92.21	86.45	89.57	**88.11**	82.95	78.48	80.65	67.57	92.55	87.62	90.02	81.85	69.71	70.72	70.21	54.10
BIT [51]	3.01	16.88	66.50	89.91	88.63	89.26	80.61	80.65	73.96	77.16	62.81	80.97	84.95	82.91	70.81	55.93	64.29	59.82	42.67
TFI-GR-Net [52]	27.78	9.74	32.02	92.09	89.60	90.83	83.20	85.12	81.69	83.37	71.49	94.70	93.16	93.92	88.54	**82.70**	73.84	78.02	63.96
DMINet [53]	6.24	14.55	64.39	92.30	88.90	90.60	82.70	82.15	79.40	80.75	67.71	59.50	83.22	69.39	53.13	71.70	63.65	67.44	50.88
USSFC [42]	1.52	4.82	23.23	89.00	91.13	90.05	81.90	75.05	81.91	78.33	64.38	88.54	**94.39**	91.37	84.11	60.57	68.06	64.10	47.17
A2Net [41]	3.78	6.02	69.43	**93.17**	89.84	91.47	84.29	85.75	79.74	82.63	70.41	93.68	93.73	93.71	88.16	75.62	74.48	75.04	60.06
GAS-Net [54]	23.52	289.97	14.39	90.98	91.07	91.02	83.53	80.65	77.77	79.18	65.54	92.47	91.33	91.90	85.01	75.19	68.05	71.44	55.57
CLAFA [55]	5.02	16.60	46.86	92.45	89.72	91.06	83.59	80.37	**84.16**	82.23	69.82	74.30	87.23	80.25	67.02	75.43	72.76	74.07	58.82
CCL-Net [56]	5.23	8.69	37.99	88.86	**91.75**	90.29	82.29	83.01	73.47	77.95	63.87	79.07	89.53	83.98	72.38	62.99	60.71	61.83	44.75
EAFH-Net(ours)	4.81	2.93	78.00	92.36	91.49	**91.74**	84.74	85.49	83.10	**84.28**	**72.83**	**96.26**	93.91	**95.07**	**90.60**	80.16	**78.28**	**79.21**	**65.57**

Algorithm 1 Flowchart of EAFH-Net

Input:

The set of image pairs T_1^n, T_2^n *from the Nth batch along with their labels* Y_L *batch size* B, *maximum training iteration* M, *and learning rate lr*

output:

four 256×256 *change maps, learned under the supervision of* Y_L.

for$[1, M]$**do:**

backbone extract feature :

$\{feat_1^{T1}, feat_2^{T1}, feat_3^{T1}, feat_4^{T1}, feat_5^{T1}\}$

$\{feat_1^{T2}, feat_2^{T2}, feat_3^{T2}, feat_4^{T2}, feat_5^{T2}\}$

After the steps in Eq.1-2 :

$\{D_2, D_3, D_4, D_5\}$

After the steps in Eq.3 :

$\{D_2', D_3', D_4', D_5'\}$

The upsampled change maps are obtained:

$C \in \{C_2, C_3, C_4, C_5\}$

The loss L_{Bce} *and* L_{Dice} *of calculatin C as Eq.19-20, The total loss is calculated by summing the two losses and then backpropagating.*

end for

experimental procedures and evaluation metrics. Finally, we delve deeply into analyzing the experimental outcomes.

A. Datasets

In an endeavor to illuminate the broad-ranging adaptability inherent in our proposed EAFH-Net model, our study entailed rigorous experimentation employing three prevalent public datasets, as well as our newly proposed UAV-CD+ dataset collected from low-altitude UAVs. These experiments were divided into two groups.

In summary, in the comparative experiments, we compared three U-Net-based methods, three recent advanced CD methods, two transformer-based derivative methods, and four lightweight CD methods.

The initial set of experiments was conducted to validate the efficacy of EAFH-Net using three widely utilized public datasets, thus demonstrating its effectiveness. The subsequent set of experiments aimed at validating the efficacy of EAFH-Net on our proposed UAV-CD+ dataset, which comprises low-altitude UAV aerial images. Furthermore, we compared our dataset with existing baselines to confirm its effectiveness. Detailed information regarding the datasets is provided below.

1) Public Datasets: To illustrate the generalization and robustness of our proposed model, we conducted experiments using LEVIR-CD [18], SYSU-CD [19], WHU-CD [20], and CLCD [21]. LEVIR-CD, containing a large number of building changes, was evaluated to demonstrate effectiveness in real-world building CD scenarios; SYSU-CD, encompassing various changes such as buildings, vegetation, and water bodies, was utilized to validate the model's effectiveness in complex change scenarios; CLCD, primarily focused on cropland changes but also including a small number of building, road, and lake changes, was employed to assess the model's sensitivity to cropland changes and minor complex scenarios. The LEVIR-CD dataset is characterized by its inclusion of 637 pairs of HR images, with each image exhibiting dimensions of 1024×1024 pixels. The images originate from diverse urban landscapes across Texas, USA, capturing temporal changes spanning from 2002 to 2018, and offer a spatial resolution of 0.5 m/pixel. Samples are cropped into 256×256 pixel patches, meticulously partitioned into training, validation, and test subsets, maintaining a ratio of 7:1:2. The SYSU-CD dataset is composed of 20 000 pairs of orthographic aerial image patches, each with dimensions of 256×256 pixels and a spatial resolution of 0.5 m. The patches are

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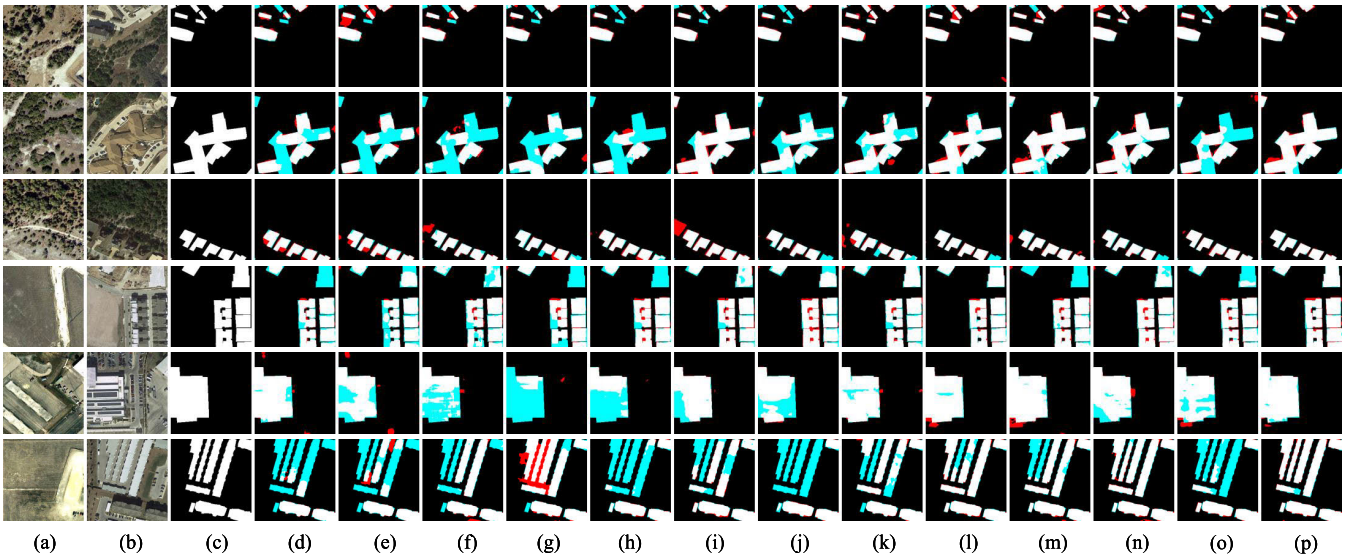


Fig. 6. Visual comparisons of the proposed method and the SOTA approaches on the LEVIR-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) FC-Siam-Diff. (e) FC-Siam-Conc. (f) SNUNet. (g) DSIFN. (h) BIT. (i) TFI-GR-Net. (j) DMINet. (k) USSFC. (l) A2Net. (m) GAS-Net. (n) CLAFA. (o) CCL-Net. (p) Ours. The colors of the images represent TPs (white), FPs (red), TNs (black), and FNs (blue).

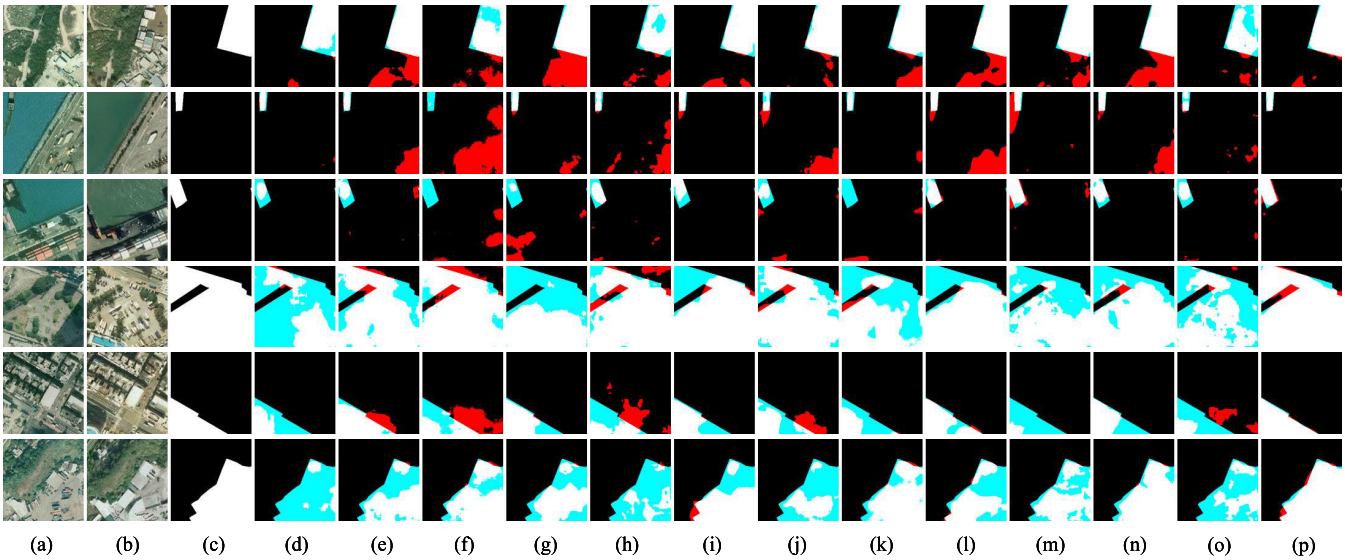


Fig. 7. Visual comparisons of the proposed method and the SOTA approaches on the SYSU-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) FC-Siam-Diff. (e) FC-Siam-Conc. (f) SNUNet. (g) DSIFN. (h) BIT. (i) TFI-GR-Net. (j) DMINet. (k) USSFC. (l) A2Net. (m) GAS-Net. (n) CLAFA. (o) CCL-Net. (p) Ours. The colors of the images represent TPs (white), FPs (red), TNs (black), and FNs (blue).

allocated across the training, validation, and test sets, maintaining an equitable distribution ratio of 6:2:2. CLCD comprises 600 pairs of agricultural land image samples, each boasting a pixel dimension of 512 × 512 pixels. The imagery was acquired within the confines of Guangdong Province, China, between the years 2017 and 2019, featuring a spatial resolution spanning from 0.5 to 2 m. Change annotations cover various features including structures like buildings, roads, and lakes, as well as barren expanses of soil, among others. The dataset is meticulously cropped into 256 × 256 pixel images, thereafter apportioned among the training, validation, and test sets following a ratio of 6:2:2. The WHU-CD dataset, proposed by Wuhan University, is a HR building CD dataset, which was collected in Christchurch, New Zealand. This dataset includes pairs of images taken in 2012 and 2016, with each image having dimensions of 32 507 × 15 354 pixels and a spatial resolution of 0.2 m/pixel. In our experiment, the overlapping areas of these images were cropped into 256 × 256 patches. The dataset was then randomly split into training, validation, and test sets in an 8:1:1 ratio, resulting in 6096 images for training, and 762 images each for validation and testing.

2) UAV-CD+ Datasets: As explained in Section III, UAV-CD+ comprises 2002 pairs of low-altitude UAV images, each featuring a resolution of 1024 × 1024 pixels and an impressive spatial resolution of 0.06 m. In the subsequent experiments, the images are cropped into several pairs of 256× 256 images. Subsequently, the dataset is divided into training, validation, and test sets, with a proportion of 8:1:1. It encompasses a diverse range of change types captured in complex

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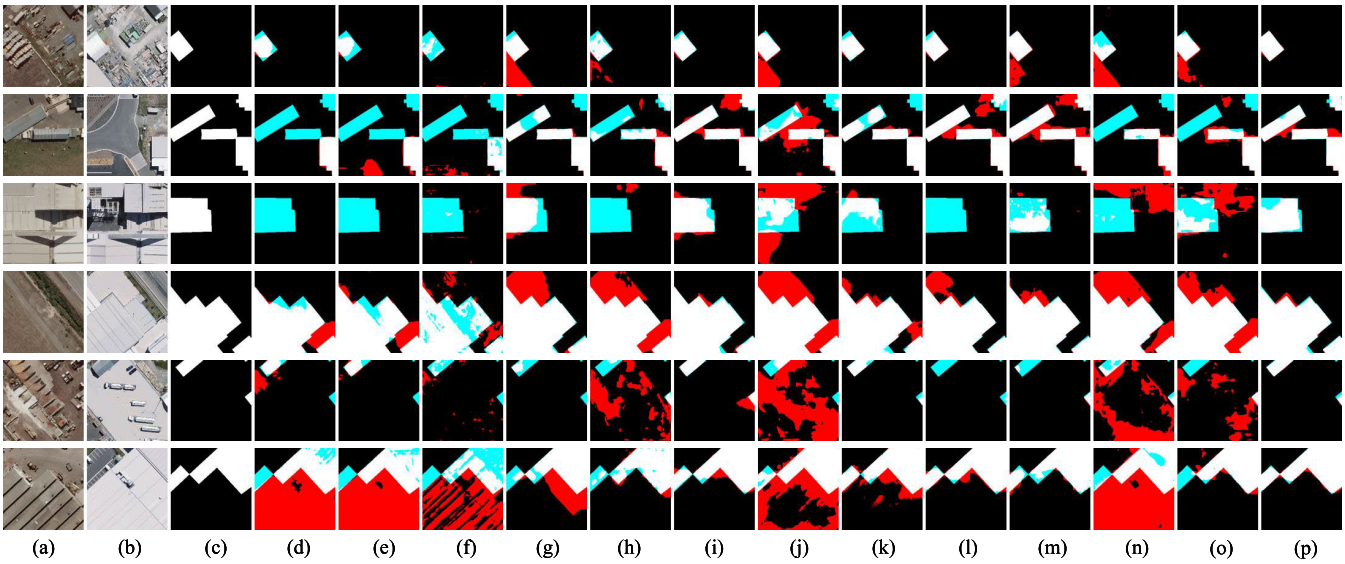


Fig. 8. Visual comparisons of the proposed method and the SOTA approaches on the WHU-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) FC-Siam-Diff. (e) FC-Siam-Conc. (f) SNUNet. (g) DSIFN. (h) BIT. (i) TFI-GR-Net. (j) DMINet. (k) USSFC. (l) A2Net. (m) GAS-Net. (n) CLAFA. (o) CCL-Net. (p) Ours. The colors of the images represent TPs (white), FPs (red), TNs (black), and FNs (blue).

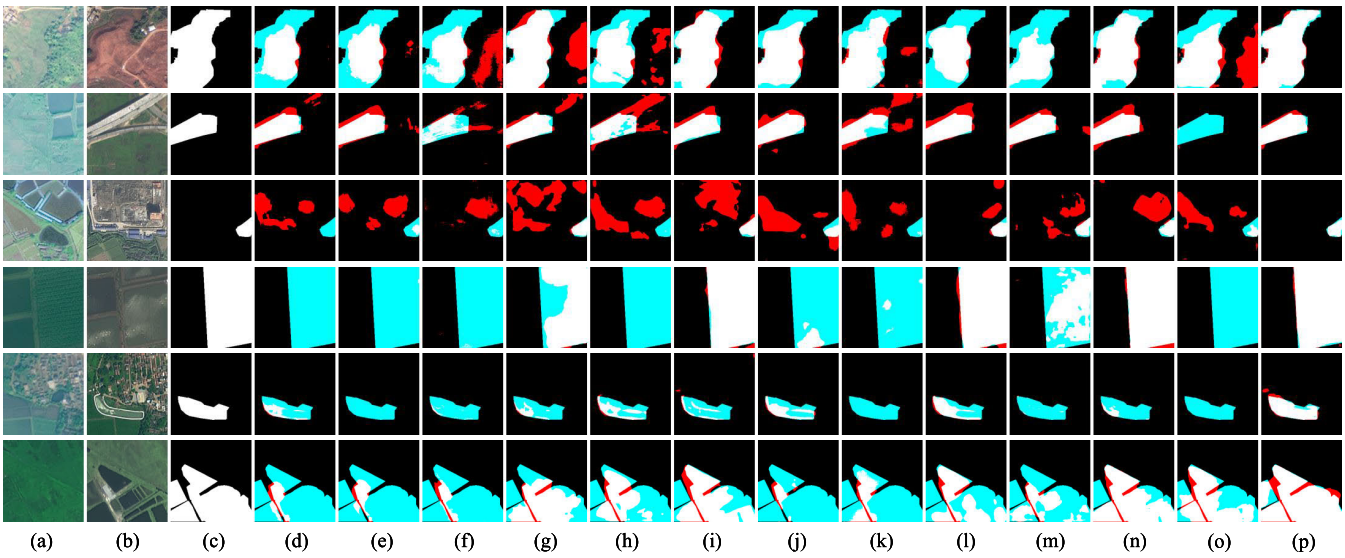


Fig. 9. Visual comparisons of the proposed method and the SOTA approaches on the CLCD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) FC-Siam-Diff. (e) FC-Siam-Conc. (f) SNUNet. (g) DSIFN. (h) BIT. (i) TFI-GR-Net. (j) DMINet. (k) USSFC. (l) A2Net. (m) GAS-Net. (n) CLAFA. (o) CCL-Net. (p) Ours. The colors of the images represent TPs (white), FPs (red), TNs (black), and FNs (blue).

scenes from low-altitude aerial photography, as visualized in Fig. 2.

B. Comparative Methods

As illustrated in Fig. 6, we provide a comparison between several recently proposed CD networks and our proposed network with regards to model $F1$-score ($F1$), parameters (Params), and FLOPs. The models subjected to comparison: FC-Siam-conc (FC-Conc) [33], FC-Siam-diff (FC-Diff) [33], SNU-Net [50], DSIFN [25], BIT [51], TFI-GR-Net [52], DMI-Net [53], USSFC [42], A2Net [41], GAS-Net [54], CLAFA [55], and CCL-Net [56].

FC-Conc and FC-Diff are trailblazers within the realm of CD, and subsequent works are compared with them. DSIFN, BIT, SNUNet, and TFI-GR-Net are, respectively, the most classic transformer-based and CNN-based CD methods in recent years. DMI-Net and GAS-Net are the latest CD methods, while USSFC, A2Net, CLAFA, and CCL-Net are lightweight methods.

1) *FC-Conc:* Diverging from the conventional FC-EF, FC-Conc diverges by incorporating dual skip connections during the decoding process. It employs a Siamese encoder to capture features from bi-temporal images and routes these features from identical levels in the encoder to their corresponding decoder counterparts.
2) *FC-Diff:* In contrast to FC-Conc, the FC-Diff module computes the absolute disparity of features extracted from corresponding tiers within the Siamese encoder, before propagating them to the decoder.

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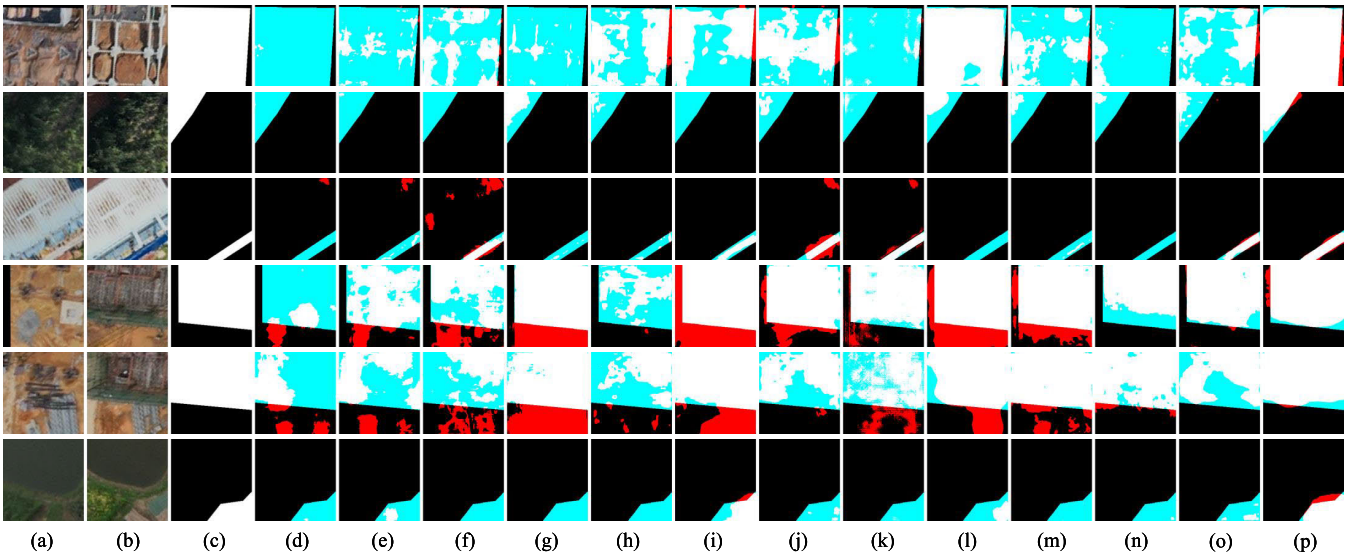


Fig. 10. Visual comparisons of the proposed method and the SOTA approaches on the UAV-CD+ dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) FC-Siam-Diff. (e) FC-Siam-Conc. (f) SNUNet. (g) DSIFN. (h) BIT. (i) TFI-GR-Net. (j) DMINet. (k) USSFC. (l) A2Net. (m) GAS-Net. (n) CLAFA. (o) CCL-Net. (p) Ours. The colors of the images represent TPs (white), FPs (red), TNs (black), and FNs (blue).

TABLE III

QUANTITATIVE ANALYSIS OF RSCD MODELS (PRECISION, RECALL, $F1$, AND IOU) ON UAV-CD+. THE BEST VALUES ARE IN BOLD. ALL RESULTS ARE REPORTED IN PERCENTAGES (%)

Method(%)	Pre	Rec	F1	IoU
FC-Siam-Diff	71.93	47.20	57.00	39.86
FC-Siam-Conc	65.22	55.32	59.86	42.72
SNUNet	61.38	41.76	49.70	33.07
DSIFN	73.37	59.86	66.01	49.27
BIT	70.76	56.93	63.09	46.09
TFI-GR-Net	64.89	**69.80**	67.26	50.67
DMINet	67.89	63.78	65.77	48.99
USSFC	63.44	65.78	64.58	47.69
A2Net	74.32	59.24	65.93	49.17
GAS-Net	68.98	63.75	66.26	49.54
CLAFA	75.18	61.59	67.71	51.18
CCL-Net	74.95	59.43	66.30	49.58
EAFH-Net(Ours)	**77.13**	64.28	**70.12**	**53.99**

3) *SNU-Net:* By employing a Siamese network with dense connections for detection, the SNU-Net tackles the challenge of deep localization information loss by effectively transmitting information not only between the encoder and decoder but also within the decoder architecture.
4) *DSIFN:* The DSIFN represents a sophisticated deep supervised image fusion network crafted specifically for HR bi-temporal RS image CD tasks. It utilizes a Siamese encoder to capture information from bi-temporal images and channels this information from identical levels in the encoder to their corresponding decoder counterparts. These deep features are subsequently propagated into a deep supervision difference discrimination network dedicated to CD.
5) *TFI-GR-Net:* The introduced CD network improves the interaction among bi-temporal features and captures temporal discrepancies across different feature levels utilizing a temporal feature interaction module. In addition, it consolidates basic and advanced temporal difference representations via a guided refinement module, aiming to enrich the positional details of advanced-level features and eliminate background noise from basic-level features.
6) *BIT:* Efficiently capturing spatio-temporal context within bi-temporal images is achieved through a bi-temporal image transformer. This model leverages a transformer encoder to succinctly model contextual information within a token-based spatio-temporal framework.
7) *DMI-Net:* This model embraces a hierarchical framework with multiple branches to efficiently and effectively generate representations of changes. The proposed method consolidates self-attention and cross-attention functionalities into a unified module, termed the cross-period joint attention block. This component orchestrates the overall feature distribution of each input, fostering the interaction between representations within individual layers and across different layers.
8) *USSFC:* Introducing a multiscale decoupled CNN method, this approach adeptly captures features of changing objects at multiple scales, particularly when combined with the widely used atrous spatial pyramid pooling (ASPP) technique.
9) *A2Net:* A novel lightweight network is proposed, which incorporates an NAM to fuse nearby features in the backbone network, a PCIM to extract temporal difference information from bitemporal features, and a SAM to aggregate high-level and low-level features.
10) *GAS-Net:* Propose a network consisting of the global-attention module (GAM) and foreground-awareness module (FAM), which both learn contextual relationships and enhance symbiotic relation learning between the scene and foreground.
11) *CLAFA:* Propose a network where the derived difference feature pyramid is progressively fused in an

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TABLE IV

ABLATION ANALYSIS OF RSCD MODELS ($F1$ AND IOU) ON FOUR DATASETS. THE BEST VALUES ARE IN BOLD. ALL RESULTS ARE DESCRIBE AS PERCENTAGES (%)

Variants	Params(M)	FLOPs(G)	LEVIR-CD		SYSU-CD		WHU-CD		CLCD		UAV-CD+	
			IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
EAFH-Net	4.81	2.93	84.74	91.74	**72.83**	**84.28**	**90.60**	**95.07**	**65.57**	**79.21**	53.99	70.12
(a) Baseline												
Baseline	2.25	0.88	66.13	79.61	54.77	70.78	68.94	81.62	47.30	64.22	37.69	54.74
(b) MFNFM												
MFNFM	4.05	1.77	83.57	91.05	71.72	83.53	88.14	93.70	63.21	77.46	46.12	63.13
All-Feature Fusion	5.34	3.99	84.44	91.57	70.90	82.97	88.44	93.87	62.33	76.80	49.13	65.89
Without concatenation	4.70	2.79	84.63	91.67	71.83	83.61	88.41	93.85	60.59	75.46	53.24	69.49
(c) SCFHM												
SCFHM	3.01	2.04	82.74	90.55	64.28	78.26	86.13	92.55	58.67	73.95	45.48	62.53
RSU7(1)	3.84	2.38	84.11	91.37	69.97	82.33	89.54	94.48	59.89	74.91	49.73	66.43
RSU7(2)	4.08	2.80	84.51	91.60	72.14	83.82	87.64	93.41	62.70	77.07	52.80	69.11
RSU4-4	4.22	2.93	**84.85**	**91.80**	72.78	84.24	88.99	94.18	64.41	78.36	51.26	67.78
RSU5-5	4.41	2.93	84.70	91.72	72.31	83.93	88.97	94.16	62.51	76.93	53.27	69.52
RSU6-6	4.6	2.93	84.59	91.66	72.37	83.97	88.33	93.81	63.88	77.96	49.06	65.82
(d)Difference backbone												
EfficientNet [59]	3.92	2.55	83.92	91.26	70.34	82.59	85.50	92.18	62.91	77.23	51.30	67.81
ShuffleNetV2 [60]	3.05	2.19	81.90	90.05	68.15	81.05	82.41	90.36	55.58	71.45	52.76	69.07
ResNet-18	13.96	6.93	83.92	91.26	72.03	83.74	88.14	93.69	62.91	77.23	53.48	69.69
ResNet-50	26.87	52.43	84.11	91.37	71.92	83.66	88.08	93.66	59.37	74.50	51.59	68.07
ResNet-101	44.06	90.54	83.94	91.27	71.60	83.45	84.25	91.45	64.64	78.52	**54.49**	**70.55**
(e)Loss												
Only BCE	-	-	84.13	91.38	71.94	83.68	89.44	94.43	65.14	78.89	53.51	69.72
Only Dice	-	-	84.36	91.52	70.69	82.83	89.87	94.67	60.78	75.61	49.83	66.52

TABLE V

LOSS CONSTRAINT ABLATION ANALYSIS OF RSCD MODELS ($F1$ AND IOU) ON FOUR DATASETS. THE BEST VALUES ARE IN BOLD. ALL RESULTS ARE DESCRIBE AS PERCENTAGES (%)

Variants	LEVIR-CD		SYSU-CD		WHU-CD		CLCD		UAV-CD+	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
constraint(λ)										
0.1	84.30	91.48	72.50	**84.06**	89.30	94.39	64.43	78.37	51.74	68.19
0.2	84.50	91.60	72.15	83.82	89.64	94.54	64.69	78.56	53.17	69.42
0.3	84.58	91.64	72.38	83.98	89.75	94.60	65.21	78.94	52.74	69.06
0.4	84.27	91.47	72.31	83.93	89.32	94.36	65.26	**78.98**	53.58	69.77
0.5	84.34	91.51	72.19	83.85	89.62	94.52	65.25	78.97	53.94	**70.08**
0.6	84.51	91.60	72.03	83.74	89.51	94.46	64.64	78.52	52.98	69.26
0.7	84.48	91.59	71.58	83.43	90.09	94.78	64.98	78.77	51.84	68.28
0.8	84.38	91.53	72.26	83.89	89.86	94.66	63.86	77.94	52.19	68.59
0.9	84.74	**91.74**	71.87	83.63	90.25	**94.88**	64.67	78.54	51.10	67.63

attend-then-filter manner, enabling high-level and adjacent lower level features to constrain each other for stable feature fusion in identifying change regions. In addition, an upsampling head is implemented with a stack of upsampling modules, each having a learnable branch that produces attentive residuals to refine statically upsampled results.

12) *CCL-Net:* Propose a cross-temporal context learning network that mines and fuses intratemporal and intertemporal long-range dependencies to exploit cross-temporal context information. Initially, a lightweight CNN extracts deep semantic features, followed by a cross-temporal fusion transformer (CFT) that locates changing objects by establishing long-range dependencies across bitemporal images.

All of the above networks have large Params and FLOPs, making them difficult to deploy for edge device CD tasks on UAVs. For example, even the smallest network, BIT, has Params and FLOPs of 3.01M and 8.48G, respectively. While achieving 89.26% detection accuracy on the LEVIR-CD dataset, its high computational cost and low accuracy make it unsuitable for high-precision and fast inference CD tasks

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on UAV edge devices. Therefore, there is a need to design a lightweight CD network with high accuracy, low FLOPS, and small Params.

C. Implementation Details

The experimental setup utilized Ubuntu 22.04.3 LTS as the operating system, an RTX 2080Ti GPU, and PyTorch [58] version 1.8.0. Parameters for the compared methods were configured as closely as possible to the original literature. In cases of insufficient memory, the batch size was adjusted to 8. EAFH-Net was initialized with pretrained weights from MobileNetV2, employing a batch size set to 32. A unified learning rate of 0.0005 was applied across all four datasets, employing the polynomial learning method

$$\left(1-\frac{\text{current iteration}}{\text{max iteration}}\right)^{\text{power}}\times \text{lr}. \tag{22}$$

The experiment proceeded for a maximum of 40 000 iterations, employing a power of 0.9 for the polynomial learning schedule. Adam optimizer [61] utilized a momentum of 0.9, a weight decay coefficient of 0.0001, and β_1 and β_2 parameters set to 0.9 and 0.99, respectively. Data augmentation strategies comprised random flipping, cropping, and temporal shuffling of input images.

To assess the efficacy of EAFH-Net, we employed five established metrics: precision (Pre), recall (Rec), $F1$, and IoU. The specific formulas for computing these metrics as outlined below

$$\text{Pre}=\frac{\text{TP}}{\text{TP}+\text{FP}} \tag{23}$$

$$\text{Rec}=\frac{\text{TP}}{\text{TP}+\text{FN}} \tag{24}$$

$$F1=\frac{2\text{Pre}\cdot\text{Rec}}{\text{Pre}+\text{Rec}} \tag{25}$$

$$\text{IoU}=\frac{\text{TP}}{\text{TP}+\text{FP}}. \tag{26}$$

True positives (TPs) represent correctly predicted positive samples, false positives (FPs) represent incorrectly predicted positive samples, true negatives (TNs) represent correctly predicted negative samples, and false negatives (FNs) represent incorrectly predicted negative samples.

D. Performance Comparison on Public Datasets and UAV-CD+

1) The performance of all methods across the three public datasets with regards to parameters, FLOPs, Pre, Rec, $F1$, and IoU are presented in Table II. Our proposed model consistently surpasses other models while maintaining the lowest computational cost among the compared methods. For instance, on the LEVIR-CD, SYSU-CD, WHU-CD, and CLCD datasets, EAFH-Net achieves $F1$ and IoU scores of 91.74% and 84.74%, 84.28% and 72.83%, 95.07% and 90.60%, and 79.21% and 65.57%, respectively, with relatively low computational costs and high FPS. EAFH-Net surpasses USSFC on all four public datasets and other models with larger parameter sizes and higher computational costs, such as FC-Diff, FC-Conc, SNU-Net, DSIFN, BIT, TFI-GR-Net, DMI-Net, A2Net, GAS-Net, CLAFA, and CCL-Net. Visual examinations of different approaches on the three public CD datasets are illustrated in Figs. 6–9.

2) Table III presents the effectiveness of each approach on the UAV-CD+ dataset in terms of Pre, Rec, $F1$, and IoU. EAFH-Net also achieves superior performance compared to other methods on the low-altitude UAV-captured dataset, with $F1$ and IoU of 70.12% and 53.99%, sequentially. Contrasted with USSFC, EAFH-Net achieves higher $F1$ and IoU. Moreover, in comparison to the approach with the highest $F1$, EAFH-Net achieves superior $F1$, suggesting that our approach achieves peak performance and accelerated computational speeds when processing high-altitude aerial imagery, RS satellite data, or low-altitude UAV-captured images. Fig. 10 provides a thorough visual comparison of all methodologies on the UAV-CD+ dataset.

E. Ablation Experiments

To thoroughly evaluate the performance of the EAFH-Net network modules, extensive ablation experiments were performed on three public datasets as well as the UAV-CD+ dataset, highlighting the influence of these modules. A comprehensive summary of the ablation experiment results across these four datasets is presented in Table IV.

1) Effectiveness of MFNFM: The efficacy of MFNFM stems from its capacity to amalgamate features extracted by neighboring sections of the backbone network, facilitating the fusion of intricate and elementary-level features across layers. This integration enhances the extracted bi-temporal information and refines granularity details. To validate its effectiveness, we replaced MFNFM with a 1×1 convolutional block, ensuring dimension alignment without compromising the network's feature extraction capability. The outcomes delineated in Table IV unequivocally underscore the performance of MFNFM.

To discuss the effectiveness of MFNFM in integrating adjacent features, we input the last four layers of features extracted by the backbone into MFNFM. To ensure a fair comparison with the original MFNFM, we perform a 3×3 convolution on these four features for dimension alignment, followed by concatenation in pairs, as shown in Fig. 12. The ablation experiment of all-feature fusion is shown in Table IV. Although the parameters and FLOPs increase, the accuracy on the LEVIR dataset shows no significant change. However, the $F1$ score on the SYSD-CD dataset drops to 82.97%, on the WHU-CD dataset drops to 93.87%, on the CLCD dataset drops to 76.80%, and on the UAV-CD+ dataset drops to 65.89%. We speculate that the aggregation of features at all scales leads to over-smoothing, resulting in the loss of local details and important boundary information. Since the LEVIR dataset primarily contains building changes, the integration of new features has little impact. In contrast, the SYSD-CD and our proposed UAV-CD+ datasets involve a wider variety of changes, making them more prone to feature redundancy when new features are introduced. In addition, the CLCD dataset, which contains many land changes, is more sensitive compared

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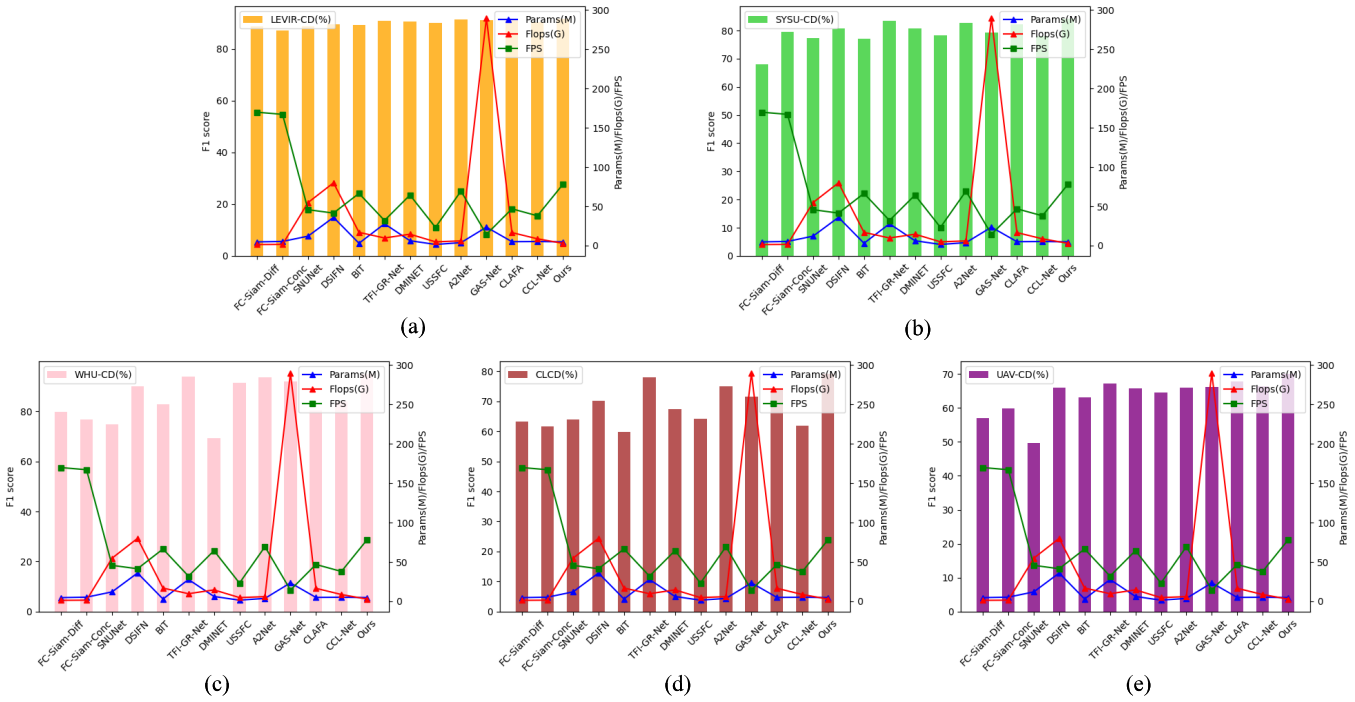


Fig. 11. Number of parameters and FLOPs of different methods and $F1$-scores of different methods on four datasets. (a) Performance on LEVIR-CD. (b) Performance on SYSU-CD. (c) Performance on WHU-CD. (d) Performance on CLCD. (e) Performance on our proposed UAV-CD+ dataset.

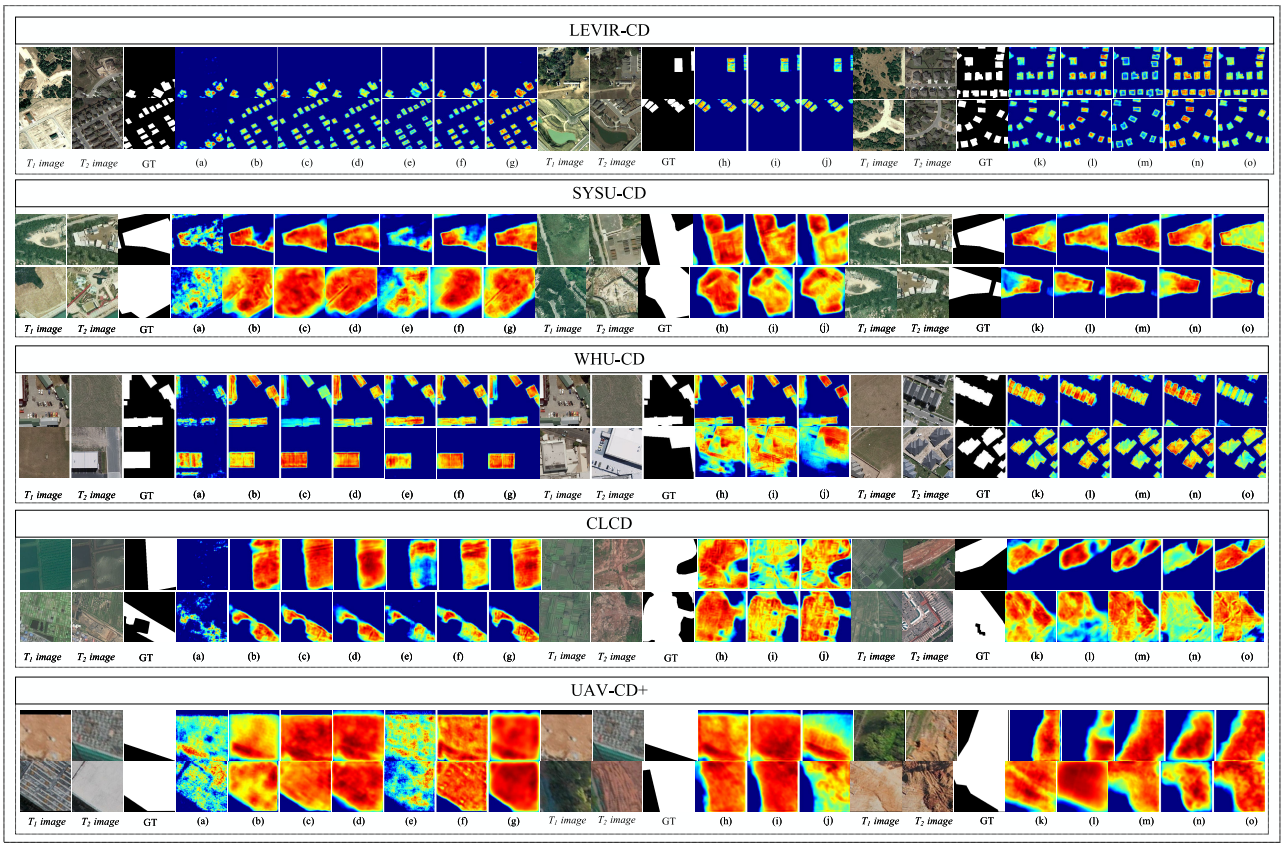


Fig. 12. Visual comparisons of the MFNFM and SCFHM and the ablation baseline modules in (a)–(g). Visual comparisons of different RSU layers in (h)–(j). Visual comparisons of different backbone in (k)–(o). (a) Visual prediction of the baseline. (b) Visual prediction without MFNFM. (c) Visual prediction with all-feature fusion module. (d) Visual prediction without concatenation. (e) Visual prediction without SCFHM. (f) Visual prediction without the RSU7(1) in SCFHM. (g) Visual prediction without the RSU7(2) in SCFHM. (h) RSU4-4. (i) RSU5-5. (j) RSU6-6. (k) EfficientNet. (l) ShuffleNetV2. (m) Resnet-18. (n) Resnet-50. (o) Resnet-101. The above are all multiscale fused feature heatmaps output from the decoder. It reflects the model's attention to areas of change.

to the other datasets, as indicated by other ablation experiments. Therefore, the over-smoothing of features significantly affects the CLCD dataset. To validate the effectiveness of the Concatenation module we added, we conducted experiments after removing the Concatenation. The results are shown in Table IV.

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2) Effectiveness of SCFHM: The SCFHM is pivotal in the harmonization and integration of spatial and channel bi-temporal features derived from the difference map. This integration accentuates the prominence of bi-temporal features within the feature space. Moreover, the insertion of an additional SCFHM between sophisticated and elementary-level features enables a more effective amalgamation of their spatial and channel bi-temporal characteristics during linking and fusion. This process culminates in the generation of more comprehensive features. Similarly, we used a 1×1 convolution to replace SCFHM, and the effectiveness of the two sets of SCFHM is shown in Table IV. In Table IV, RSU7(1) and RSU7(2) refer to the ablation test settings where the first and second sets of RSU blocks are removed from SCFHM, respectively. RSU4-4, RSU5-5, and RSU6-6 indicate the number of feature extraction repetitions in the two sets of RSU blocks. This setup is designed to demonstrate whether the depth of RSU blocks affects the final CD accuracy.

3) Discussion on the Convolutional Layer of SCFHM Module: The depth of layers in the convolutional layer of SCFHM directly influences its spatial and channel feature harmonization ability, as well as the size of parameters and computational costs. Consequently, we delve into the implications of the convolutional layer's depth within SCFHM on network performance. We conducted this experiment on three widely recognized public datasets, as shown in Table IV. We found that when the number of layers decreases, there is a small fluctuation in the increase or decrease of parameters and computational costs, along with a slight decrease in accuracy. However, when both sets of SCFHM have seven convolutional layers, their performance is more stable.

4) Discussion on the Impact of Different Backbones on the Network: We further validate different backbone networks fused with our proposed method, such as ResNet18, ResNet50, and ResNet101. The results from Table IV demonstrate that using MobileNetV2 as the backbone network, along with a slight decrease in accuracy parameter count and computational cost are lower than ResNet18, ResNet50, and ResNet101, by 9.15M, 22.06M, and 39.25M, respectively, with corresponding reductions in FLOPs of 4G, 49.5G, and 90.54G. On LEVIR-CD, it exhibits $F1$ improvements of 0.48%, 0.37%, and 0.47%; on SYSU-CD, improvements of 0.54%, 0.62%, and 0.83%; on CLCD, improvements of 1.98%, 4.71%, and 0.69%; on UAV-CD+, improvements of 0.43% and 2.05%. However, when used as a backbone network on UAV-CD+, its $F1$ is lower by 0.43% compared to ResNet101. Nonetheless, as a lightweight network architecture, it still achieves commendable performance. This demonstrates that MobileNetV2 can attain excellent performance while ensuring network lightweightness. Since our proposed method emphasizes lightweight design, we incorporated ablation experiments focused on lightweight backbones. In these experiments, we replaced the EAFH-Net backbone with EfficientNet and ShuffleNetV2. The experimental results indicate that EfficientNet and ShuffleNetV2 exhibit weaker fitting capabilities for RS bitemporal data compared to MobileNetV2.

5) Discussion on Constraint Conditions of Hybrid Loss Function on the Network: To verify the effectiveness of the hybrid loss function, we conducted ablation experiments on cross-entropy loss and dice loss separately. The implementation data is shown in Table IV. From the ablation experiments, it can be observed that the hybrid use of cross-entropy loss and dice loss performs the best across the five datasets. In addition, we added constraints to the dice loss, with the constraints and experimental details provided in Table V.

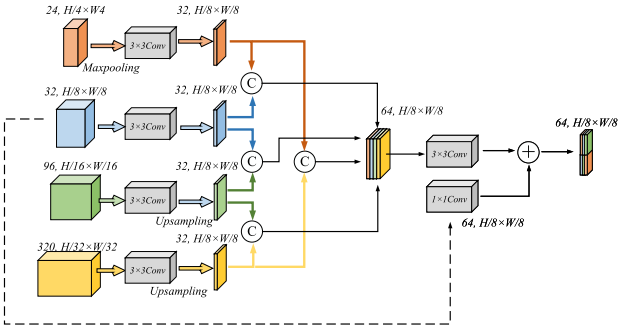


Fig. 13. Overview of MFNFM (all-feature fusion version).

VI. Conclusion

In this article, this article introduces EAFH-Net, a novel lightweight CD network, along with UAV-CD+, a benchmark dataset captured by low-altitude drones. UAV-CD+ significantly enhances existing CD datasets by addressing deficiencies in pixel resolution, spatial resolution, change types, and dataset volume for low-altitude captured data. EAFH-Net comprises an MFNFM module and two sets of SCFHM modules. MFNFM facilitates the fusion of features extracted by neighboring extraction of the backbone network, enabling cross-fusion of sophisticated and elementary-level features, thereby enhancing bi-temporal information extraction and refining granularity details. The first set of SCFHM harmonizes and integrates spatial and channel bi-temporal features of the difference map, accentuating the prominence of bi-temporal features in the feature space. The other set of SCFHM, inserted between sophisticated and elementary-level features, enhances the combination of their spatial and channel bi-temporal features during linking and fusion, resulting in more comprehensive features. Experimental results demonstrate that EAFH-Net achieves faster computational speed and superior performance on three public CD datasets and UAV-CD+, illustrating its adaptability to RS satellite images, high-altitude aerial images, and UAV-captured datasets, thereby showcasing the generality and robustness of our proposed network. In future endeavors, we will further optimize the dataset's change variety and the network's complexity and computational speed.

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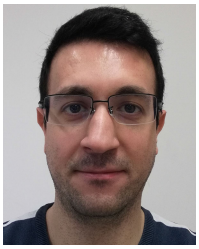
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